HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F78PB

Probability and Statistics B

Semester 2 2017/18

Duration: Two Hours

ANSWER ALL QUESTIONS

Total Marks: 60

Provided:
Distributions Formula Sheet
Correlation and Regression Formula Sheet
Yellow Actuarial Tables
Blue Statistical Tables

1. A company is being bothered by nuisance phone calls. It measures the time in hours between nuisance calls over a period of days. The data are shown in Table 1.1

6.2	4.0	8.7	1.7	5.5	4.1	10.1	11.0	7.7	7.2
1.9	6.7	8.4	7.5	6.5	6.8	0.6	21.1	4.9	15.4

Table 1.1

Note:

$\sum x = 146,\ \sum x^2 = 1484.76,\ \sum (x - \bar{x})^3 = 2511.876$

(a) Calculate the mean, standard deviation and the coefficient of symmetry.

(5 marks)

(b) It is thought that these data come from a Gamma(3, θ) distribution. The Maximum Likelihood Estimate for θ is given by $\frac{n\bar{X}}{(n-1)S^2}$ (you do not need to prove this). Estimate the value of θ for these data.

(1 mark)

(c) For the Gamma(a, b) distribution, the coefficient of symmetry is approximately equal to $2/\sqrt{a}$. Verify this for the data in Table 1.1 and by using sketches explain what the coefficient of symmetry measures.

(2 marks)

(d) The kurtosis for the data in Table 1.1 calculates as 1.62. Explain what this means.

(2 marks)

[Total 10 marks]

2. Suppose that $\underline{X} = (X_1, X_2, ..., X_n)$ are independent and identically distributed random variables from a distribution with parameter θ and density function:

$$f(x) = \begin{cases} \frac{2x}{\theta^2} & 0 \le x \le \theta \\ 0 & \text{otherwise} \end{cases}$$

Let $g(\underline{X})$ denote an estimator for θ.

(a) Give definitions for the *bias* and *mean square error (MSE)* of $g(\underline{X})$.

(2 marks).

(b) For $g(\underline{X}) = X_{(n)} = \max\{X_1, X_2, ..., X_n\}$ show that the cumulative distribution function is given by

$$F_{X_{(n)}}(x) = \left(\frac{x}{\theta}\right)^{2n} \quad \text{for} \quad 0 \le x \le \theta$$

(6 marks)

(c) From your answer to (b) derive the probability density function of the estimator $X_{(n)}$.

(3 marks)

(d) Show that the estimator $\frac{2n+1}{2n} X_{(n)}$ is an *unbiased* estimator for θ.

(3 marks)

(e) It can be shown that $MSE\left(\frac{2n+1}{2n} X_{(n)}\right) = \frac{\theta^2}{4n^2(n+1)}$. Why does this make $\frac{2n+1}{2n} X_{(n)}$ a *consistent* estimator?

(1 mark)

[Total 15 marks]

3. For an independent and identically distributed random sample of size n from the Beta(θ,1) distribution (the density is given by $f(x;\theta) = \theta x^{\theta-1}$ for $0 < x < 1$), find

(a) The method of moments estimate (MME) of θ.

(3 marks)

(b) The maximum likelihood estimate (MLE) of θ.

(6 marks)

[Total 9 marks]

4. The data below are the scores of 19 randomly chosen candidates in a psychometric test for an accountancy internship.

79	90	101	83	72	76	71	93	91	87
76	86	80	89	77	69	61	86	81	

Table 4.1

Note: $\sum x = 1548$ and $\sum (x - \bar{x})^2 = 1670.737$

(a) By calculating the appropriate numerical statistics, construct a boxplot for the data in Table 4.1.

(4 marks)

(b) Explain why $h(\underline{X},\mu) = \frac{\bar{x} - \mu}{s/\sqrt{n}}$ is a pivotal quantity for μ, the population mean of all candidates taking the test.

(3 marks)

(c) Calculate a 95% confidence interval for the population mean based on the sample results. Justify any assumptions you make in obtaining this interval.

(3 marks)

(d) The HR department claims that the average score for all candidates taking this psychometric test is 85. Does this seem a reasonable claim? Use (c) to justify your answer.

(2 marks)
[Total 12 marks]

5. A researcher wishes to investigate whether a coin is fair or not, that is if $P(heads) = 0.5$. She throws the coin four times and decides to accept the hypothesis $H_0 : P(heads) = 0.5$ in a test against the alternative $H_1 : P(heads) \neq 0.5$, if the number of times that the coin lands "heads" is 1, 2, or 3.

(a) Calculate the probability of the type I error of this test.

(3 marks)

(b) Calculate the probability of the type II error of this test, if the true probability that the coin lands "heads" is 0.7.

(3 marks)
[Total 6 marks]

6. A random sample of ten claim amounts and the corresponding payment amounts on household contents policies is taken from a particular insurance company. The table below gives the claim amounts and payment amounts (both in £1000) for each of the claims considered.

Claim x :	2.80	3.40	3.60	5.00	5.80	6.20	6.80	7.00	7.60	8.60
Payment y :	3.06	2.82	3.78	3.92	6.04	5.20	6.74	6.12	7.46	7.60

For these data $\sum x_i = 56.8$, $\sum x_i^2 = 356.4$, $\sum y_i = 52.74$, $\sum y_i^2 = 306.786$, $\sum x_i y_i = 329.364$

(a) Calculate the sample correlation coefficient for these data and interpret what this value tells you.

(4 marks)

(b) We want to test the hypothesis that the true value of the correlation coefficient, ρ, is greater than 0.9. Determine a suitable null and alternative hypothesis and perform this test using a statistic based on Fisher's Z transformation.

(4 marks)
[Total 8 marks]

END OF PAPER